CSCI218: Foundations of Artificial Intelligence

UNIVERSITY
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What is AI?

AI = making intelligent machines

Standard model: machines are intelligent to the extent that their actions can be expected to achieve their objectives

AI comprises:

- problem solving, constraint satisfaction, games
- knowledge representation, reasoning, planning
- natural language processing
- speech, vision, robotics
- machine learning

The goal is general-purpose AI: capable of quickly learning high-quality behavior in “any” task environment

What if we succeed?

- Lift the living standards of everyone on Earth to a respectable level
  - => 10x increase in world GDP ($13.5Q net present value)
- Potential advances in health, education, science

- Massive disruption of human economic roles
- Need to resolve the issues related to safety and ethics

Reasons to doubt

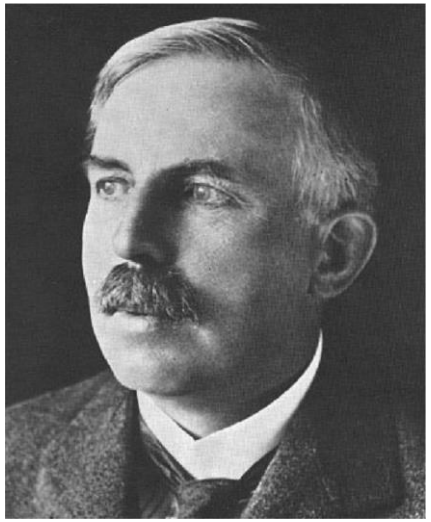
- The rapid progress of recent years does not prove that there can be no limits to what AI can achieve
- Alan Turing (1950), the first person to define AI, was also the first to raise possible objections to AI.
- The argument from informality (human behavior is too complex)
- The argument from disability (A machine can never do *X*)
- The mathematical objection (Incompleteness theorem)

Towards human-level AI

Still missing:

- Real understanding of language
- Integration of learning with knowledge
- Long-range thinking at multiple levels of abstraction
- Cumulative discovery of concepts and theories

Date unpredictable



Sept 11, 1933: Lord Rutherford addressed BAAS:
"Anyone who looks for a source of power in the transformation of the atoms is talking moonshine."

Sept 12, 1933: Leo Szilard invented the neutron-induced nuclear chain reaction

Eventually (but not very soon)…

AI systems will make better real-world decisions than humans

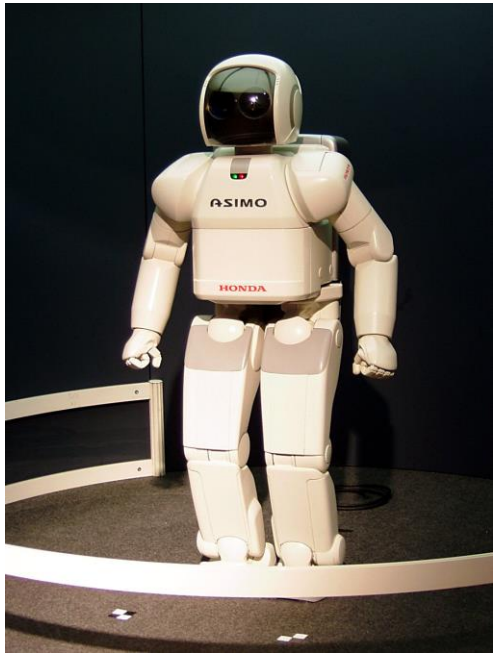
Turing's point: how do we retain power over entities more powerful than us, for ever?

-- "It seems probable that once the machine thinking method had started, it would not take long to outstrip our feeble powers," ***he said during a lecture in 1951****.*

"At some stage therefore, we should have to expect the machines to take control."

Can Machines Really Think?

- Some philosophers claim that a machine that acts intelligently would not be *actually* thinking, but would be only a *simulation* of thinking
- Most AI researchers are not concerned with the distinction
  - Whether submarine can swim?
  - Whether airplanes can fly?



The Ethics of AI

- AI is a powerful technology. We have a moral obligation to use it well, to promote the positive aspects and avoid or mitigate the negative ones.
- Positive aspects
  - Medical diagnosis, extreme weather prediction, safer driving, food production;
  - Natural disaster recovery, pollution monitoring, suicide prevention, etc.
  - Many more to name
- Negative
  - Economic and societal impact, increased income inequality, autonomous weapon systems, large-scale surveillance, data security, privacy concerns, etc.

The Ethics of AI

- **Ethics examines the rational justification for our moral judgments; it studies what is morally right or wrong, just or unjust**

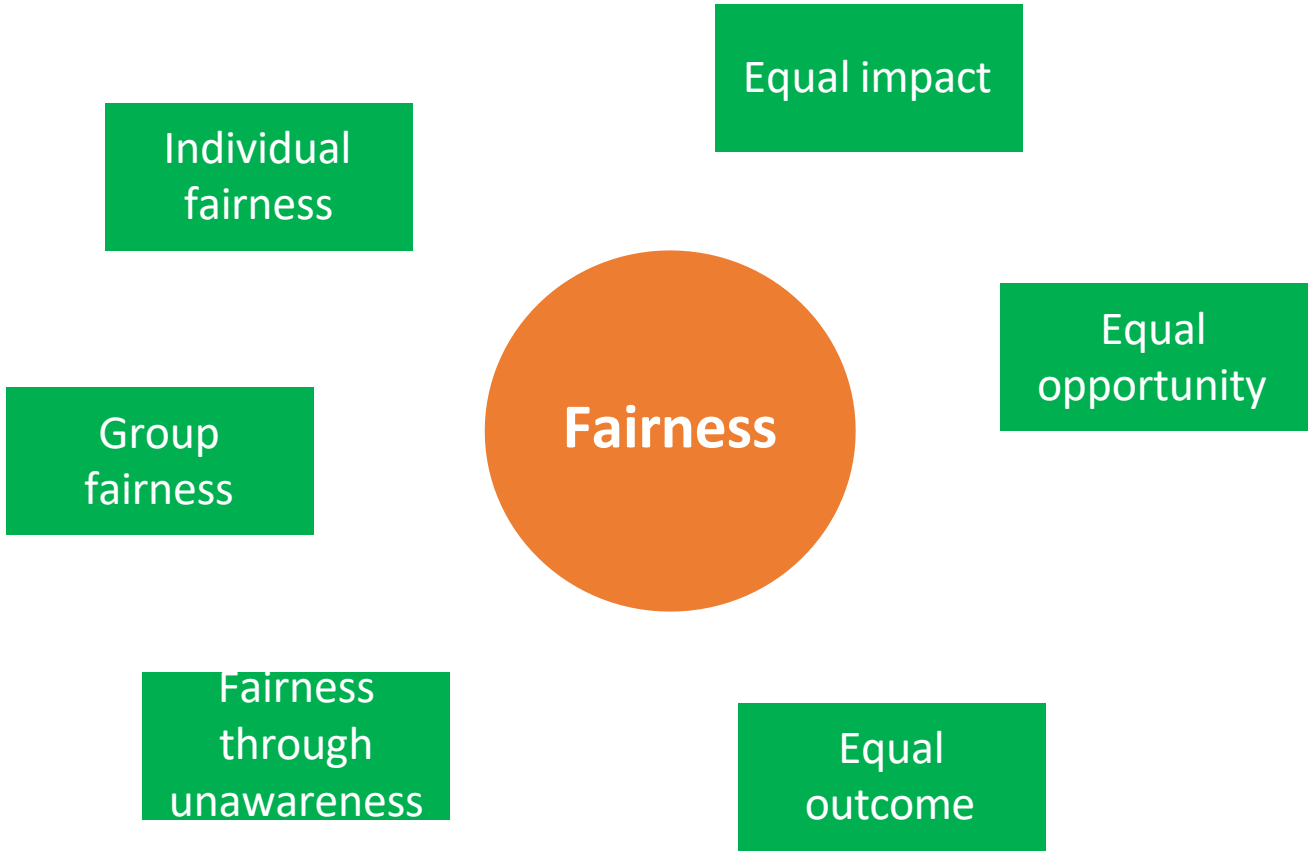
The Ethics of AI

- **All scientists and engineers face ethical considerations**
  - **of what projects they should or should not take on,**
  - **how they can make sure the execution of the project is safe and beneficial.**
- **Every organization and everyone has a responsibility to make sure the technology contributes to good, not harm.**

Ensure safety	Establish accountability
Ensure fairness	Uphold human rights and values
Respect privacy	Reflect diversity/inclusion
Promote collaboration	Avoid concentration of power
Provide transparency	Acknowledge legal/policy implications
Limit harmful uses of AI	Contemplate implications for employment

Fairness and Bias

- **Six of the most commonly-used concepts**



Fairness and Bias

- Machine learning is making decisions in important situations
  - Whose loan gets approved, to what neighborhoods police officers are deployed, who gets pretrial release or parole.
- But machine learning models can perpetuate societal bias.
  - A system picks up the (racial or gender) prejudices of human judges from the examples in the training set
  - Designers of machine learning systems have a moral responsibility to ensure that their systems are in fact fair.
- Sample size disparity can also lead to biased results
  - fewer training examples of minority class than of majority class.

Fairness and Bias

- We need to defend against the biases
  - Understand the limits of the data we are using
  - Train engineers to be aware of issues of fairness and bias
  - Debias the data (e.g., minority over-sampling technique)
  - Invent models and algorithms that are more resistant to bias
  - Debias the recommendations of an initial model

Images are from https://www.geektime.com/understanting-ml-bias-fairness/

Fairness and Bias

- How to make sure that the systems we build will be fair?
  - Talk with social scientists and domain experts to understand the issues and perspectives, and consider fairness from the start.
  - Create an environment that fosters a diverse pool of software engineers.
  - Define what groups your system will support: different language speakers, different age groups, different abilities with sight and hearing, etc.
  - Optimize for an objective function that incorporates fairness.
  - Examine your data for prejudice and for correlations between protected attributes and other attributes.
  - Understand how any human annotation of data is done.
  - Make sure you track metrics for subgroups that might be victims of bias.
  - Include system tests that reflect the experience of minority group users.
  - Ensure that when fairness problems come up, they are dealt with.

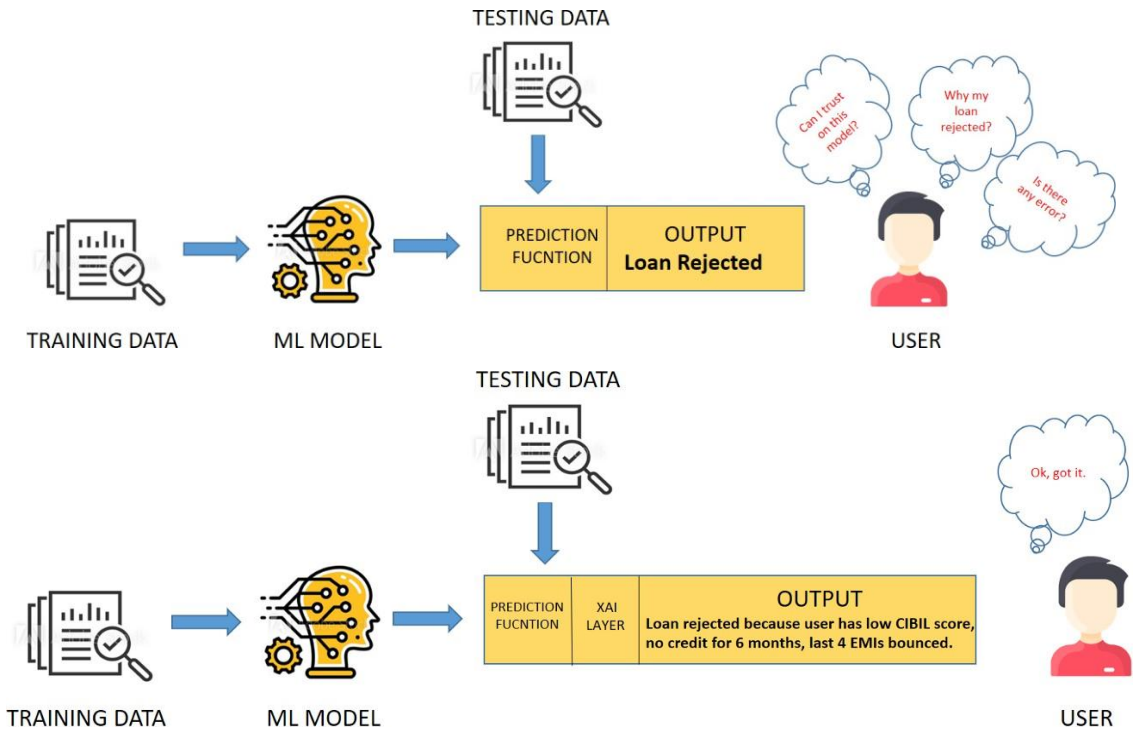
Trust and transparency

- People need to be able to trust the systems they use
- Verification and validation (V&V) process
  - Verification means that the product satisfies the specifications.
  - Validation means ensuring that the specifications actually meet the needs of the user and other affected parties.
- We need to verify
  - The data that these systems learn from;
  - The accuracy and fairness of the results;
  - Adversaries cannot unduly influence the model, nor steal information by querying the resulting model.
- One instrument of trust is certification
  - The AI industry is not yet very clear on safety standards, although there are some frameworks in progress

Trust and transparency

- Another aspect of trust is transparency
  - Consumers want to know what is going on inside a system, and that the system is not working against them
- Explainable AI (XAI)
  - A set of processes and methods that allows human users to comprehend and trust the results and output created by machine learning algorithms.
- A good explanation has several properties:
  - Be understandable and convincing to the user,
  - Accurately reflect the reasoning of the system,
  - Be complete, and
  - Be specific in that different users with different conditions or different outcomes should get different explanations.

Trust and transparency



- **We say that a system is interpretable if we can inspect the source code of the model and see what it is doing**
- **We say it is explainable if we can make up a story about what it is doing—even if the system itself is an uninterpretable black box.**

https://medium.com/analytics-vidhya/explainable-ai-the-next-level-c6b4dadc240

AI Safety

- AI agents are deployed in safety-critical applications
  - Driving cars, controlling robots, and making life-or-death medical decisions.
- It would be unethical to distribute an unsafe AI agent.
- We require our agents to avoid accidents, to be resistant to adversarial attacks and malicious abuse, and in general to cause benefits, not harms.
- Safety goes beyond to that the specification has considered any feasible failure modes, and is designed to degrade gracefully even in the face of unforeseen failures.

Standard model of AI

Machines whose actions can be expected to achieve their objectives

But we cannot specify objectives completely and correctly

(Third wish = please undo first two wishes)

We shall avoid unintended side effects

Maintain low impact, making cost part of the utility function

Example: Video game

Several agents in video games discovered ways to crash or pause the game when they were about to lose, thus avoiding a penalty.

And in a specification where crashing the game was penalized, one agent learned to use up just enough of the game's memory so that when it was the opponent's turn, it would run out of memory and crash the game.

- **With incompletely or incorrectly defined objectives, better AI => *worse* outcomes**

- **Shall not maximize utility without actually solving the problem that their designers intended them to solve.**

A new model

- ~~Machines are intelligent to the extent that their actions can be expected to achieve their objectives~~
- Machines are ***beneficial*** to the extent that **their** actions can be expected to achieve ***our*** objectives

Provably Beneficial AI

1. **Robot goal: satisfy human preferences**
2. **Robot is *uncertain* about human preferences**
3. **Human behavior provides evidence of preferences**

The robot solves a formally defined **assistance game**

Optimal solutions:

defer to human, ask permission, allow self to be switched off

It is rational for humans to build machines that solve assistance games

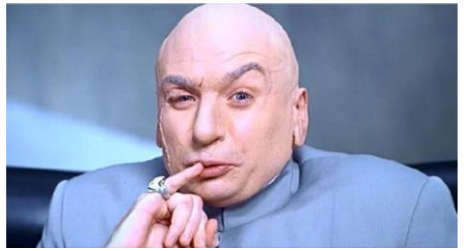
Better AI => better outcomes

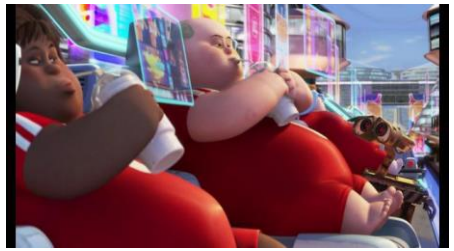
Summary and implications

AI has vast potential and unstoppable momentum

Provably beneficial AI is possible *and desirable*

Problems of misuse and overuse remain open





Acknowledgement

The lecture slides are based on the materials from ai.Berkey.edu

Questions?

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